James (Yoon Jae) Nam

jamesnam12345@gmail.com
https://jamesnam12345.github.io/

Summary

Senior Software Engineer with over 12 years of experience building scalable real-time communication systems and mobile/web infrastructure. Spent 9 years at Google as a core contributor and engineering leader on the Google Chat team. Expert in cross-platform architecture, concurrency, and full-stack engineering. Passionate about applying AI to solve complex engineering and user problems.

Professional Experience

Member of Technical Staff | **Armadin** (Palo Alto, CA) | **April 2026 – July 2026**

- Implemented backend read/write APIs for core asset management operations, using Spanner, Python, and Protocol Buffers.
- Enhanced the Armadin AI attacker validation system to support more parameters and improved resource efficiency.
- Built three new Grafana dashboards on Prometheus/GCP metrics, streamlining issue diagnosis for the engineering team.
- Developed a new, low-latency cost-alerting pipeline to keep the team quickly informed of cost related issues.
- Built an LLM-driven network graph simulation framework to evaluate GNNs against classical baselines, such as Gradient Boosted Tree and Adamic-Adar, for vulnerability inference. Also explored using GNNs with MCTS to guide AI attacker strategy.

Senior Software Engineer | **Google** (Sunnyvale, CA) | **October 2016 – January 2026**

AI Innovation & Leadership

- **AI Prototypes**: Developed multiple full-stack AI prototypes using Firebase Studio and Antigravity, including a natural language CRM interface, a multimodal (video+audio) agent for e-commerce, and a Chat application that uses Gemini function calling.
- **AI Debugging Tools**: Built an AI-driven debugging tool using context-aware prompt engineering to diagnose crash reports.
- **Mentorship**: Managed 10 interns, with 7 returning as full-time. Conducted 100+ interviews and served on hiring committees.
- **Awards**: Awarded 10+ peer bonuses and Gold Awards for contributions to multiplatform engineering and concurrency.

Core Infrastructure & Architecture

- **Cache Invalidation Protocol**: Architected a revision-based cache invalidation protocol for resolving data integrity issues.
- **Web on Shared**: Led core parts of the "Web on Shared" initiative, unifying data layer across Android, iOS, and Web.
- **Kotlin Multiplatform (KMP)**: Contributed to KMP efforts via 10+ regex-based static analysis checks to prevent regressions.
- **Concurrency & Stability**: Reduced iOS crash rates by diagnosing concurrency issues and authoring thread-safety guidelines.

Product Scale & Feature Growth

- **Scale & Growth**: Played a pivotal role in scaling the product over 10x, enabling Guest Access and Consumer Access.
- **User Capabilities**: Designed the "Chat User Capabilities" system, a centralized permission engine adopted across 20+ features.
- **Sync Reliability**: Resolved critical user-visible sync issues by enhancing the revisioned sync mechanism, ensuring data freshness.

Software Engineer, Growth team | **Coursera** (Mountain View, CA) | **January 2015 – September 2016**

- Built key workflows (Home, Dashboard, Onboarding) and analyzed 100+ A/B tests.

Software Engineer | **Palantir Technologies** (Palo Alto, CA) | **January 2014 – January 2015**

- Re-architected a major product into a single-page application (SPA), significantly improving user experience.

Education

Stanford, CA | **Stanford University** | **2010 – 2014**

- B.S.E. in Computer Science (conferred in January 2014)
- Awards: Frederick E. Terman Award (Top 5% of Engineering Majors)
- GPA: Major 4.1/4.3, Overall 4.0/4.3.

Certifications

- Deep Learning Specialization (Coursera / DeepLearning.AI)
- Machine Learning Specialization (Coursera / Stanford Online, DeepLearning.AI)

Skills

- **Languages**: Java, Python, Go, Swift, Kotlin, Objective-C, JavaScript/TypeScript, SQL, C++
- **Technologies**: Gemini API, AI Agents, Flutter, React, Django, Next.js, Firebase, Tensorflow, PyTorch, Gemini CLI, Antigravity, Spanner, Prometheus, Grafana, LangGraph, NumPy, Terraform
- **Concepts**: Real-time Communication, Distributed Systems, Concurrency, Cross-Platform Development (KMP), Graph Neural Networks